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FA23 -BCE-113

Numerical Computations
Lab-Quiz

“1/3 Simpson Method for Numerical Integration”

Q # 1: Implement the 1/3 Simpson's Rule in MATLAB to numerically evaluate the definite integral of a piecewise function.

$$f(x)=0 \text{ for } 0 \leq x < 30$$

$$f(x)=-9.1688 \times 10^{-6}x^3+2.7961 \times 10^{-3}x^2-0.28487x+9.6778 \text{ for } 30 \leq x \leq 100$$

$$f(x)=0 \text{ for } x > 100$$

$$I = \int (-9.1688 \cdot 10^{-6}x^3 + 2.7961 \cdot 10^{-3}x^2 - 0.28487x + 9.6778) dx \quad [\text{from } 30 \text{ to } 100]$$

where $n = 2, 4, 6, 8, 10, 12, 50$

Matlab code:

```
%% Define the piecewise function
function y = f(x)
y = zeros(size(x));
% piece 1: 0 <= x < 30, f(x) = 0
piece1 = (x >= 0) & (x < 30);
y(piece1) = 0;
% Region 2: 30 <= x <= 100, f(x) = cubic polynomial
piece2 = (x >= 30) & (x <= 100);
y(piece2) = -9.1688e-6 * x(piece2).^3 + 2.7961e-3 * x(piece2).^2 - ...
0.2848700 * x(piece2) + 9.6778;
% Region 3: 100 < x, f(x) = 0
piece3 = (x > 100);
y(piece3) = 0;
end
%% Applying 1/3 simpson rule
clear; clc; close all;
%% Integration limits
a = 0;
b = 100;
%% Different N values
N_values = [2,4,6,8,10,12,50];
fprintf('\n===== \n');
fprintf(' SIMPSON 1/3 RULE RESULTS \n');
fprintf('===== \n \n');
simp13_results = [];
for N = N_values
if mod(N, 2) ~= 0
N = N + 1; % Ensure N is even
end
```

```

h = (b - a) / N;
x = linspace(a, b, N+1);
y_vals = f(x);
sum_odd = sum(y_vals(2:2:N));
sum_even = sum(y_vals(3:2:N-1));
I_simp13 = (h/3) * (y_vals(1) + 4*sum_odd + 2*sum_even + y_vals(end));
simp13_results(end+1,:) = [N h I_simp13];
end
%% ---- EXACT ANALYTICAL INTEGRAL ----
syms X
f_exact = -9.1688e-6*X^3 + 2.7961e-3*X^2 - 0.28487*X + 9.6778;
I_exact = double(int(f_exact, 30, 100));
fprintf('Exact Integral = %.10f\n\n', I_exact);
%% ---- PRINT TABLE ----
fprintf('-----\n');
fprintf(' N h Simpson I Error\n');
fprintf('-----\n');
for i = 1:size(simp13_results,1)
N = simp13_results(i,1);
h = simp13_results(i,2);
I_simp = simp13_results(i,3);
err = abs(I_exact - I_simp);
fprintf('%4d %10.6f %.10f %.6e\n', ...
N, h, I_simp, err);
end
fprintf('-----\n');

```

Output:

```

=====
SIMPSON 1/3 RULE RESULTS
=====

```

Exact Integral = 60.7926153333

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N	h	Simpson 1/3	Error
2	50.000000	84.9466666667	2.415405e+01
4	25.000000	26.9166666667	3.387595e+01
6	16.666667	66.6063786008	5.813763e+00
8	12.500000	62.3177864583	1.525171e+00
10	10.000000	85.8201866667	2.502757e+01
12	8.333333	58.4138065844	2.378809e+00
50	2.000000	65.4213528149	4.628737e+00

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